AI-Powered FOD Detection System

in Space Exploration

PROBLEM

Foreign Object Debris (FOD) is a serious safety risk in:
Spacecraft assembly areas
Laboratories and clean rooms
Industrial production zones
Launch and testing environments
Small objects such as cables, screws or tools can cause:
Equipment damage
Mission failure
Financial losses
Safety hazards
Current detection methods are manual and unreliable.

Solution

We developed an AI-powered monitoring system that:

- Automatically detects foreign objects
- Classifies the risk level
- Generates safety recommendations
- Works in real time

The system is automated, scalable and low-cost.

Product

System architecture:

- 4 ESP32-CAM cameras
- Central Python processing unit
- AI analysis via API

Workflow:

Camera → Python → AI Analysis → Risk Report → Alert

Technology Stack

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Technology Stack

Hardware	Software	Output:
• ESP32-CAM • Raspberry Pi / PC / Jetson	• Python • OpenCV • HTTP requests • AI API integration	• JSON reports • Image snapshots • Risk classification

Results

Current prototype includes:

- 4-camera monitoring system
- Automatic image capture
- AI detection pipeline
- Structured safety reports

Example result:

Detected object: cable

Risk level: medium

Recommended action: remove object from workspace

Use Cases

Space Industry:

- Satellite assembly
- Space station maintenance

Industrial Production:

- Electronics manufacturing
- Robotics workshops

Research Labs:

- Universities
- Clean rooms

Value

The system provides:

- Automated safety monitoring
- Reduced human error
- Prevention of equipment damage
- Increased operational safety
- Cost-effective solution

Future Development

Next steps:

- ROS2 integration
- Autonomous inspection robot
- Local AI model (edge inference)
- Real-time alert system
- Integration with security systems

Thank you for your attention